

Not Finished, But Abandoned

by

Cameron T. Kuchta

A dissertation submitted in partial fulfillment of
the requirements for the degree of

Doctor of Philosophy

(Physics)

at the

UNIVERSITY OF WISCONSIN–MADISON

2026

Date of final oral examination: 01/01/2100

The dissertation is approved by the following members of the Final Oral Committee:

Jane Doeverything, Professor, Electrical Engineering

John Dosomethings, Associate Professor, Electrical Engineering

© Copyright by Cameron T. Kuchta 2026
All Rights Reserved

ACKNOWLEDGMENTS

It is customary for authors of academic books to include in their prefaces statements such as this: "I am indebted to ... for their invaluable help; however, any errors which remain are my sole responsibility." Occasionally an author will go further. Rather than say that if there are any mistakes then he is responsible for them, he will say that there will inevitably be some mistakes and he is responsible for them....

Although the shouldering of all responsibility is usually a social ritual, the admission that errors exist is not — it is often a sincere avowal of belief. But this appears to present a living and everyday example of a situation which philosophers have commonly dismissed as absurd; that it is sometimes rational to hold logically incompatible beliefs.

— DAVID C. MAKINSON (1965)

Above is the famous "preface paradox," which illustrates how to use the `wbepi` environment for epigraphs at the beginning of chapters. You probably also want to thank the Academy.

CONTENTS

Contents ii

List of Tables v

List of Figures vi

Abstract vii

1 Introduction 1

1.1	<i>What the heck is a plasma</i>	1
1.1.1	Plasmas in the universe	1
1.1.2	How do plasmas affect us on earth?	1
1.1.3	Where does magnetic reconnection play into this whole shabang?	1
1.2	<i>Magnetic Reconnection</i>	1
1.2.1	The Generalized Ohm's Law	1
1.2.1.1	How do we derive this?	1
1.2.1.2	A breakdown of the terms	1
1.2.2	Regimes of Reconnection	1
1.2.2.1	Reconnection scaling parameters	1
1.2.2.2	Where do various systems line up?	2
1.2.2.3	What are we interested in?	2
1.2.3	Sweet-Parker Reconnection	2
1.2.3.1	A quick historical detour	2
1.2.3.2	What are the assumptions that go into Sweet-Parker?	2
1.2.3.3	Sweet-Parker Reconnection Rate Derivation	2
1.2.3.4	Application to Real Systems	2

1.2.3.4.1	What are some example parameters where reconnection occurs? .	2
1.2.3.4.2	What time scales does Sweet-Parker predict?	2
1.2.3.4.3	What did we miss?	2
1.2.4	Hall Reconnection	2
1.2.4.1	Why should the Hall term appear?	2
1.2.4.1.1	What is the relevant physics that sets when the hall term matters .	2
1.2.4.1.2	In what kind of plasmas does the hall term appear and when does it not?	2
1.2.5	Kinetic Reconnection	3
1.2.5.1	What makes kinetic reconnection “kinetic”? .	3
1.2.5.2	Where does Pressure Anisotropy come from? .	3
1.2.5.2.1	Movement of flux tube	3
1.2.5.2.2	Effect of background field	3
1.2.5.3	How does pressure anisotropy change what the reconnection looks like	3
1.2.5.3.1	Thinner reconnection layer?	3
1.2.5.3.2	Marginal Firehose Condition	4
1.2.5.3.3	Concluding: what features should we look for in the data?	4
1.2.5.4	Why do we want to measure it?	4
1.3	<i>Experimental Setup</i>	4
1.3.1	Overall how does our experiment progress	4
1.3.2	Initial field	4
1.3.2.1	Helmholtz	4
1.3.2.2	Permanent magnets	4
1.3.2.3	Toroidal	4

1.3.2.3.1	Shifted coil	4
1.3.2.3.2	Axial coil	4
1.3.2.4	Perturbation coils	4
1.3.3	Plasma Generation	4
1.3.3.1	How do our guns work	4
1.3.3.1.1	Gas lines	4
1.3.3.1.2	PFNs	4
1.3.3.1.3	Puff trigger	4
1.3.3.1.4	How to breakdown	5
1.3.3.2	How are the guns placed for injection? . .	5
1.3.3.3	What goes wrong with them	5
1.3.3.3.1	Burning up	5
1.3.3.3.2	Burn holes in wall because of back-ground field	5
1.3.3.3.3	Air cooling	5
1.3.3.3.4	Optimizing gun operation	5
1.3.4	Drive field	5
1.3.4.1	Individual coils	5
1.3.4.1.1	Zone's between coils as plasma sources	5
1.3.4.2	Drive Cylinder	5
1.3.5	Diagnostics Overview	6
1.3.6	How does this compare to space	6
2	Pressure Wave	7
3	Multi-tip Langmuir Probe	8
4	Pressure Anisotropy Probe	9
5	Conclusion	10
	Colophon	11

LIST OF TABLES

1.1	List of diagnostics used in the experiment and what they can measure.	5
1.2	Comparison of lab and space parameters.	6

LIST OF FIGURES

1.1	Diagram of reconnection regimes.	1
1.2	Diagram of Sweet-Parker reconnection.	2
1.3	Diagram of 2d hall reconnection	2
1.4	Flux tube changing size during reconnection causing pressure anisotropy build up.	3
1.5	Simulated kinetic reconnection with changing guide field. . .	3
1.6	Diagram from Ohia of with and without pressure anisotropy in simulation.	3
1.7	Some figure from Jan's papers thowing effects of pressure anisotropy	3
1.8	Circuit diagram of plasma gun.	4
1.9	Simplified cut of plasma gun insides.	5
1.10	Image from high speed camera of gun spewing	5
2.1	Greess et al. 2022 paper showing simulations with pressure wave.	7

ABSTRACT

FIXME: basically a placeholder; do not believe

I did some research, read a bunch of papers, published a couple myself, (pick one):

1. ran some experiments and made some graphs,
2. proved some theorems

and now I have a job. I've assembled this document in the last couple of months so you will let me leave. Thanks!

1 INTRODUCTION

1.1 What the heck is a plasma

1.1.1 Plasmas in the universe

1.1.2 How do plasmas affect us on earth?

1.1.3 Where does magnetic reconnection play into this whole shabang?

1.2 Magnetic Reconnection

1.2.1 The Generalized Ohm's Law

1.2.1.1 How do we derive this?

1.2.1.2 A breakdown of the terms

1.2.2 Regimes of Reconnection

1.2.2.1 Reconnection scaling parameters

i.e. why do we use S and L/d_i to characterize reconnection regimes?

Figure 1.1: Diagram of reconnection regimes.

Figure 1.2: Diagram of Sweet-Parker reconnection.

Figure 1.3: Diagram of 2d hall reconnection

1.2.2.2 Where do various systems line up?

1.2.2.3 What are we interested in?

1.2.3 Sweet-Parker Reconnection

1.2.3.1 A quick historical detour

1.2.3.2 What are the assumptions that go into Sweet-Parker?

1.2.3.3 Sweet-Parker Reconnection Rate Derivation

1.2.3.4 Application to Real Systems

1.2.3.4.1 What are some example parameters where reconnection occurs?

1.2.3.4.2 What time scales does Sweet-Parker predict?

1.2.3.4.3 What did we miss?

1.2.4 Hall Reconnection

1.2.4.1 Why should the Hall term appear?

1.2.4.1.1 What is the relevant physics that sets when the hall term matters

1.2.4.1.2 In what kind of plasmas does the hall term appear and when does it not?

Figure 1.4: Flux tube changing size during reconnection causing pressure anisotropy build up.

Figure 1.5: Simulated kinetic reconnection with changing guide field.

Figure 1.6: Diagram from Ohia of with and without pressure anisotropy in simulation.

1.2.5 Kinetic Reconnection

1.2.5.1 What makes kinetic reconnection “kinetic”?

1.2.5.2 Where does Pressure Anisotropy come from?

1.2.5.2.1 Movement of flux tube

1.2.5.2.2 Effect of background field

1.2.5.2.2.1 Too low of background field does...

1.2.5.2.2.2 Too high of field does...

1.2.5.2.2.3 Goldilocks!

1.2.5.3 How does pressure anisotropy change what the reconnection looks like

1.2.5.3.1 Thinner reconnection layer? Use results from Jan’s more recent papers.

Figure 1.7: Some figure from Jan’s papers showing effects of pressure anisotropy

Figure 1.8: Circuit diagram of plasma gun.

1.2.5.3.2 Marginal Firehose Condition

1.2.5.3.3 Concluding: what features should we look for in the data?

1.2.5.4 Why do we want to measure it?

1.3 Experimental Setup

1.3.1 Overall how does our experiment progress

1.3.2 Initial field

1.3.2.1 Helmholtz

1.3.2.2 Permanent magnets

1.3.2.3 Toroidal

1.3.2.3.1 Shifted coil

1.3.2.3.2 Axial coil

1.3.2.4 Perturbation coils

1.3.3 Plasma Generation

1.3.3.1 How do our guns work

1.3.3.1.1 Gas lines

1.3.3.1.2 PFNs

1.3.3.1.3 Puff trigger

Figure 1.9: Simplified cut of plasma gun insides.

Figure 1.10: Image from high speed camera of gun spewing

Table 1.1: List of diagnostics used in the experiment and what they can measure.

1.3.3.1.4 How to breakdown

1.3.3.2 How are the guns placed for injection?

1.3.3.3 What goes wrong with them

1.3.3.3.1 Burning up

1.3.3.3.2 Burn holes in wall because of background field Get image from Paul with hole in wall.

1.3.3.3.3 Air cooling

1.3.3.3.4 Optimizing gun operation

1.3.4 Drive field

1.3.4.1 Individual coils

1.3.4.1.1 Zone's between coils as plasma sources

1.3.4.2 Drive Cylinder

Reference Paul's RSI + thesis.

Table 1.2: Comparison of lab and space parameters.

1.3.5 Diagnostics Overview

1.3.6 How does this compare to space

2 PRESSURE WAVE

Figure 2.1: Greess et al. 2022 paper showing simulations with pressure wave.

3 MULTI-TIP LANGMUIR PROBE

4 PRESSURE ANISOTROPY PROBE

5 CONCLUSION

DISCARD THIS PAGE

COLOPHON

This template uses Gyre Pagella by default. (I used Arno Pro in my dissertation.)

Feel free to give me a shout-out in your colophon or acks if this template is useful for you. Good luck!